

Foundations of Stable Diffusion

[Stable Diffusion]

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

1.0 Content Block

Getty Images asserted that the training and development of Stable Diffusion involved the unauthorized use of its images, which were downloaded on servers and computers that were potentially in the UK. However, Stability AI argued that all training and development took place outside the UK, specifically in U.S. data centers operated by Amazon Web Services. Stability AI applied for reverse summary judgment and/or strike out of two claims: the training and development claim, and the secondary infringement of copyright claim. The High Court, however, refused to strike out these claims, allowing them to proceed to trial. The court is to determine whether the training and development of Stable Diffusion occurred in the UK, which is crucial for establishing jurisdiction under the UK's Copyright, Designs and Patents Act 1988 (CDPA). The secondary infringement claim revolved around whether the pre-trained Stable Diffusion software, made available in the UK through platforms like GitHub, HuggingFace, and DreamStudio, constitutes an "article" under sections 22 and 23 of the CDPA. As of November 4, 2025 Getty Images largely lost the lawsuit over AI image generator to Stability AI, prompting Getty and some lawyers to call for stronger protections for copyright owners in Britain.

2.0 Content Block

User interfaces Stability provides an online image generation service called DreamStudio. The company also released an open source version of DreamStudio called StableStudio. In addition to Stability's interfaces, many third party open source interfaces exist, such as AUTOMATIC1111 Stable Diffusion Web UI, which is the most popular and offers extra features, Fooocus, which aims to decrease the amount of prompting needed by the user, and ComfyUI, which has a node-based user interface, essentially a visual programming language akin to many 3D modeling applications.

3.0 Content Block

An "embedding" can be trained from a collection of user-provided images, and allows the model to generate visually similar images whenever the name of the embedding is used within a generation prompt. Embeddings are based on the "textual inversion" concept developed by researchers from Tel Aviv University in 2022 with support from Nvidia, where vector representations for specific tokens used by the model's text encoder are linked to new pseudo-words. Embeddings can be used to reduce biases within the original model, or mimic visual styles. A "hypernetwork" is a small pretrained neural network that is applied to various points within a larger neural network, and refers to the technique created by NovelAI developer Kurumuz in 2021, originally intended for text-generation transformer models. Hypernetworks steer results towards a particular direction, allowing Stable Diffusion-based models to imitate the art style of specific artists, even if the artist is not recognised by the original model; they process the image by finding key areas of importance such as hair and eyes, and then patch these areas in secondary latent space. DreamBooth is a deep learning generation model developed by researchers from Google Research and Boston University in 2022 which can fine-tune the model to generate precise, personalised outputs that depict a specific subject, following training via a set of images which depict the subject.

4.0 Content Block

Training procedures The model was initially trained on the laion2B-en and laion-high-resolution subsets, with the last few rounds of training done on LAION-Aesthetics v2 5+, a subset of 600 million captioned images which the LAION-Aesthetics Predictor V2 predicted that humans would, on average, give a score of at least 5 out of 10 when asked to rate how much they liked them. The LAION-Aesthetics v2 5+ subset also excluded low-resolution images and images which LAION-5B-WatermarkDetection identified as carrying a watermark with greater than 80% probability. Final rounds of training additionally dropped 10% of text conditioning to improve Classifier-Free Diffusion Guidance. The model was trained using 256 Nvidia A100 GPUs on Amazon Web Services for a total of 150,000 GPU-hours, at a cost of \$600,000.

5.0 Content Block

The text to image sampling script within Stable Diffusion, known as "txt2img", consumes a text prompt in addition to assorted option parameters covering sampling types, output image dimensions, and seed values. The script outputs an image file based on the model's interpretation of the prompt. Generated images are tagged with an invisible digital watermark to allow users to identify an image as generated by Stable Diffusion, although this watermark loses its efficacy if the image is resized or rotated. Each txt2img generation will involve a specific seed value which affects the output image. Users may opt to randomize the seed in order to explore different generated outputs, or use the same seed to obtain the same image output as a previously generated image. Users are also able to adjust the number of inference steps for the sampler; a higher value takes a longer duration of time, however a smaller value may result in visual defects. Another configurable option, the classifier-free guidance scale value, allows the user to adjust how closely the output image adheres to the prompt. More experimentative use cases may opt for a lower scale value, while use cases aiming for more specific outputs may use a higher value. Additional text2img features are provided by front-end implementations of Stable Diffusion, which allow users to modify the weight given to specific parts of the text prompt. Emphasis markers allow users to add or reduce emphasis to keywords by enclosing them with brackets. An alternative method of adjusting weight to parts of the prompt are "negative prompts". Negative prompts are a feature included in some front-end implementations, including Stability AI's own DreamStudio cloud service, and allow the user to specify prompts which the model should avoid during image generation. The specified prompts may be undesirable image features that would otherwise be present within image outputs due to the positive prompts provided by the user, or due to how the model was originally trained, with mangled human hands being a common example.

6.0 Content Block

End-user fine-tuning To address the limitations of the model's initial training, end-users may opt to implement additional training to fine-tune generation outputs to match more specific use-cases, a process also referred to as personalization. There are three methods in which user-accessible fine-tuning can be applied to a Stable Diffusion model checkpoint: